WHAT IS PROGRAMMING LOGIC?

Programming logic is the systematic way of thinking to solve problems

It focuses on **how to solve**, not on any language

Every good programmer first develops strong logic

Logic is the foundation of:

- Python
- Java
- C/C++
- JavaScript
- AI/ML programming

Example:

Cooking recipe = Logic
Actual cooking = Coding

WHY PROGRAMMING LOGIC IS IMPORTANT

Makes problem-solving easier

Reduces coding errors

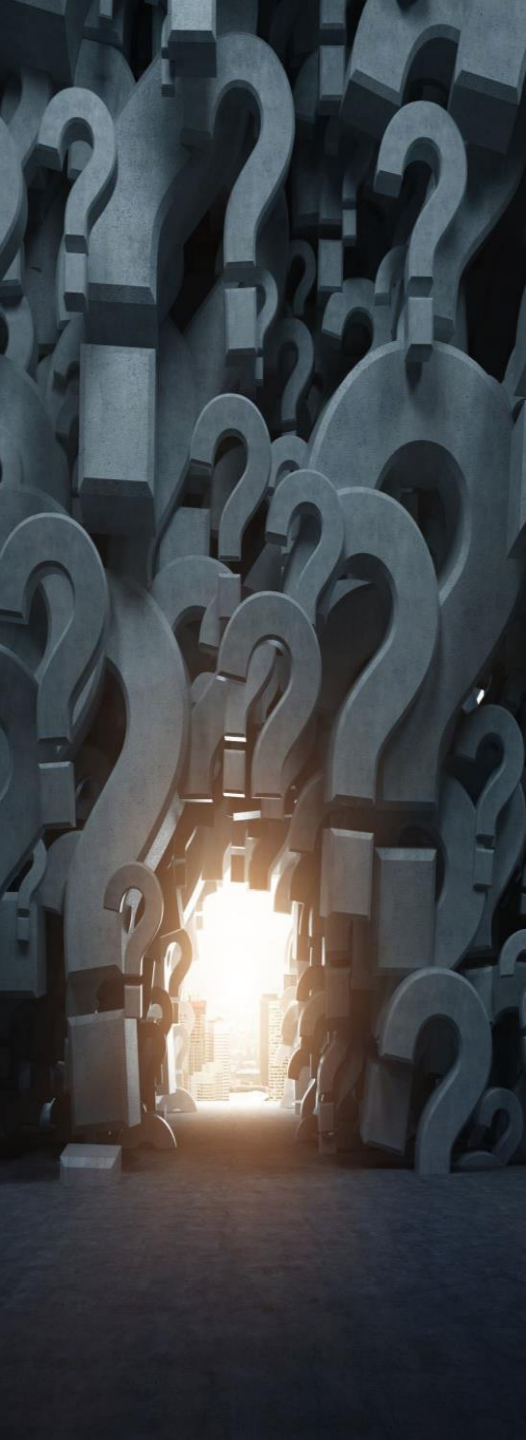
Improves confidence in interviews

Helps in competitive programming

Essential for:

- **Software development**
- **Data Science**
- **AI/ML**
- **Cybersecurity**
- **App development**

Without logic, coding becomes memorization.



PROBLEM SOLVING PROCESS

- **Every programming problem follows these steps:**
- **Understand the problem**
- **Identify inputs and outputs**
- **Break problem into smaller parts**
- **Design logic (algorithm)**
- **Represent logic (pseudocode/flowchart)**
- **Convert logic into code**
- **Test and debug**

REAL LIFE EXAMPLE OF LOGIC

Problem: Withdraw money from ATM

Logical Steps:

- Insert ATM card
- Enter PIN
- Select transaction type
- Enter amount
- Collect cash
- Take receipt
- Exit

This is exactly how a program works.

WHAT IS AN ALGORITHM?

- **An algorithm is a finite sequence of steps to solve a problem**
- **Must be:**
  - **Clear**
  - **Unambiguous**
  - **Finite**
  - **Logical**
  - **Formula:**
    Problem → Algorithm → Program → Output

EXAMPLE ALGORITHM (ADD TWO NUMBERS)

Problem: Add two numbers

Algorithm:

- Start
- Input A, B
- Sum = A + B
- Display Sum
- Stop

CHARACTERISTICS OF A GOOD ALGORITHM

- **Input: Takes input values**
- **Output: Produces meaningful result**
- **Definiteness: Every step is clear**
- **Finiteness: Ends after finite steps**
- **Effectiveness: Easy to execute**

WHAT IS PSEUDOCODE?

Pseudocode is a simple English-like representation of logic

Not a real programming language

Easy for humans to understand

Acts as bridge between algorithm and code

Used in:

Exams

Interviews

Project planning

Competitive programming

PSEUDOCODE EXAMPLE (EVEN OR ODD)

```
START
INPUT number
IF number MOD 2 = 0
 PRINT "Even Number"
ELSE
 PRINT "Odd Number"
END
```

WHAT IS A FLOWCHART?

A flowchart is a graphical representation of logic

Uses symbols to show flow of execution

Helps visualize program structure

Commonly used in:

Teaching

Documentation

System design

COMMON FLOWCHART SYMBOLS

Symbol	Shape	Purpose
Oval	Start/End	Begin and terminate
Parallelogram	Process	Calculation
Rectangle	Input/Output	Read/Display data
Diamond	Decision	Yes/No condition
Arrow	Flow line	Direction of logic

VARIABLES IN PROGRAMMING LOGIC

- **Variable = container to store data**

  Examples:

- **age = 20**
- **name = "Ali"**
- **marks = 85.5**

Why variables matter:

- **Store information**
- **Perform operations**
- **Control program behavior**

DATA TYPES (BASIC UNDERSTANDING)

- **Integer: 10, 25, -4**
- **Float: 3.14, 99.99**
- **Character: 'A', 'B'**
- **String: "Hello", "India"**
- **Boolean: True / False**

OPERATORS IN PROGRAMMING LOGIC

Operators perform actions on data.

Arithmetic Operators

+ _ * / %

Relational Operators

< == != >= <=

Logical Operators

AND, OR, NOT

Example:

IF age > 18 AND citizen = True

CONTROL STRUCTURES

Control how program runs:

- **Sequence – Step by step execution**
- **Selection – IF, IF-ELSE (decision making)**
- **Looping – FOR, WHILE (repetition)**

Example:

If marks ≥ 40 → Pass

Else → Fail